Δ proof why variance of Uniform distribution is $Var(X) = \frac{(b-a)^2}{12}$

Let X be continuos random variable $\xrightarrow{\text{then}}$ $Var(X) = E(X^2) - (E(X))^2$

Consider $\mu = E(x)$. Let X have probability density function f_X.

[probability density function (f_X): The probability density function of a continous random variable X with support S is an integrable function $f_X(x)$ satisfying the 2 conditions: 1 $\forall x \in S$, $f(x) > 0$ 2 The area under the curve $f(x)$ in the support S is 1, $\int_S f_X(x)\,dx = 1$ 3 the probability of X that $X \in A$, is $\rightarrow P(X) = \int_A f_X(x)\,dx$]

First we will prove the variance formula $\Rightarrow$ $\boxed{\mu = E(x) = \int_{-\infty}^{\infty} x f_X(x)\,dx}$

$$Var(X) = E((X-\mu)^2) = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)\,dx = \int_{-\infty}^{\infty} (x^2 - 2\mu x + \mu^2) f_X(x)\,dx$$

$$= \underbrace{\int_{-\infty}^{\infty} x^2 f_x(x)\,dx}_{E(x^2)} - \underbrace{2\mu \underbrace{\int_{-\infty}^{\infty} x f_X(x)\,dx}_{\mu}}_{-2\mu^2} + \mu^2 \underbrace{\int_{-\infty}^{\infty} f_X(x)\,dx}_{=1}$$

$$= E(x^2) - 2\mu^2 + \mu^2 = E(x^2) - \mu^2 = E(x^2) - (E(x))^2 \quad (*)$$

Δ So for uniform distribution we know that $E(x) = \frac{a+b}{2}$

$\rightarrow$ $Var(X) = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx - (E(X))^2$ $\rightarrow$ we have that $a \leqslant x \leqslant b$ so $\rightarrow$

$$Var(X) = \underbrace{\int_{-\infty}^{a} 0x^2\,dx}_{\text{out of range so } p(x)=0} + \int_a^b \frac{x^2}{b-a}\,dx + \underbrace{\int_b^{\infty} 0x^2\,dx}_{\text{out of range so } p(x)=0} - \left(\frac{a+b}{2}\right)^2$$

$$\Rightarrow Var(X) = \int_a^b \frac{x^2}{b-a}\,dx - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{x^3}{3(b-a)}\Big|_a^b - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{b^3}{3(b-a)} - \frac{a^3}{3(b-a)} - \frac{a^2+2ab+b^2}{4} = \frac{\overbrace{b^3 - a^3}^{(b-a)(a^2+ab+b^2)}}{3(b-a)} - \frac{a^2+2ab+b^2}{4}$$

$$= \frac{4\cancel{(b-a)}(a^2+ab+b^2)}{12\cancel{(b-a)}} - \frac{(a^2+2ab+b^2)}{4} = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12}$$

$$= \frac{b^2-2ab+a^2}{12} = \frac{(b-a)^2}{12}$$ ▨ we prove!